6. Non Uniform flow

In this flow, the magnitude and directions of velocities of fluid particles are not equal at all sections. Example: Constant discharge through a variable diameter pipe.

7.3.2 Energy possessed by a fluid body

Energy is the ability to do work. It is available in different forms. The energy possessed by a flowing fluid consists of energies due to position, pressure and velocity. They are classified as,

1. Potential energy or static energy or Datum energy.
2. Pressure energy.
3. Kinetic energy.

1. Potential energy

This is the energy possessed by a mass of fluid by virtue of its position in space.

2. Pressure energy

It is the energy possessed by a fluid particle by virtue of its existing pressure.

3. Kinetic energy

It is the energy possessed by a moving fluid particle by virtue of its motion or velocities.

4. Total energy

The total energy of a particle of a flowing fluid is the sum of its potential energy, pressure energy and kinetic energy.

7.3.3 Bernoulli's Theorem / Energy equation

Bernoulli's theorem states that "for a steadily flowing incompressible fluid, in which there is a continuous connection between all the particles of the fluid, the total energy or total head of each particle is the same, provided there is no loss or gain of energy while the particle moves from one point to another".

Mathematically,

$$Z + \frac{v^2}{2g} + \frac{P}{\gamma} = \text{Constant}$$

Where,

Z = Potential energy or Datum head

$\frac{v^2}{2g}$ = Kinetic energy, (v = velocity of flow, g = acceleration due to gravity)

$\frac{P}{\gamma}$ = Pressure energy, (P = Pressure, γ = unit weight of fluid)

7.4 Flow Through Orifice and Mouthpiece

7.4.1 Orifice – Definition

An opening provided in the side wall or base of a vessel, through which liquid is discharged, is called an Orifice. It is used to measure the discharge.

7.4.2 Types of Orifice

1. According to size
   - a. Small Orifice
   - b. Large Orifice
2. According to shape
   - a. Circular Orifice
   - b. Rectangular Orifice
   - c. Triangular Orifice

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